

Matching the Lens to the Zero

An exact optimal shift for the Li-Sekatskii coefficients, measured against the Weil window

Mohamed Osman | ORCID 0009-0004-5912-999X | independent researcher

August 2026

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Summary

The generalized Li criterion of **Sekatskii (2013, 2014)** — whose coefficients are now called the **Li-Sekatskii coefficients** in the literature — replaces Li’s map by

$$q_a(\rho) = (\rho - a) / (\rho - (1 - a)), \quad a \text{ real, } a \neq 1/2,$$

whose unit circle is exactly the critical line, and shows that the non-negativity of the associated sums — equivalently, of certain derivatives of $\log \xi$ evaluated at $s = 1 - a$ — is again equivalent to the Riemann Hypothesis, for every admissible a . The construction below is his; §1 names the source for each part.

This note adds a measurement and the small piece of analysis around it: **the shift is a lens with an optimal width, and the optimum has a closed form.**

For a hypothetical zero off the line at $\rho = 1/2 + \beta + iT$, write $d = 1/2 - a$ for the lens width. The exponential rate at which that zero announces itself in the shifted sums is

$$\begin{aligned} \log |q_a(\rho)| &= \frac{1}{2} \cdot \log[((\beta+d)^2 + T^2) / ((\beta-d)^2 + T^2)] \\ &= 2\beta d / (d^2 + T^2) + O(\beta^3), \end{aligned}$$

This has an **exact** maximum. Writing $R = |\rho - 1/2| = \sqrt{(\beta^2 + T^2)}$, the rate is stationary at $d = R$ and only there, and its value is $\operatorname{artanh}(\beta/R)$. Li's classical choice $a = 0$ is the lens $d = 1/2$; for $\beta \ll T$ the two rates are β/T and β/T^2 .

The lens is optimal when its foci sit at the same distance from $1/2$ as the zero does. Matching it improves the exponential detection rate — and hence the index at which the zero first shows — by a factor T .

§3.4 shows why the detection index should scale like $1/\log R$, so that maximising the rate minimises the index: the on-line terms are $1 - \cos(n\theta)$ with $\theta = 2 \arctan(\gamma/d) - \pi$, and the far zeros contribute a positive $n^2 d^2$ term. Those are the two ingredients a Palojärvi-type interval needs; the uniform estimates that would make it a certified interval are not supplied here.

§6 asks the same question of the truncated Weil form and gets the opposite answer: its support length has **no** interior optimum, the ceiling $e^{\beta L/2}$ being monotone by Paley-Wiener. The Li parameter is a position and admits a matching condition; the Weil parameter is a bandwidth and is limited by a price rather than an optimum.

The gain is real and it is measurable. It is also, on its own, worth very little: §5 explains why a better detector is not a shorter proof, and §4 records a truncation artifact — already identified analytically by Sekatskii — that will manufacture a false detection if the prime side is cut before the cancellation is done.

Status legend

tag	meaning
[K]	Known. Re-derived here for self-containedness; source named in §1. No novelty claimed.
[P]	Proved here.
[M]	Measured, with parameters stated.
[C]	Caveat.

1. Prior art

component used here	where it already lives
positivity of $\lambda_n \iff \text{RH}$	Li , <i>J. Number Theory</i> 65 (1997) 325–333
the general multiset theorem behind it	Bombieri & Lagarias , <i>J. Number Theory</i> 77 (1999) 274–287
the shifted map q_a and the derivative object $L_n(a)$	Sekatskii , arXiv:1304.7895 = <i>Ukrainian Math. J.</i> 66 (2014) 415–431 — both appear verbatim in the abstract
the equality between the zero sum and the derivative	same paper
the prime side as a generalized Laguerre transform of Λ	Sekatskii , arXiv:1404.7276, Lemma 1 and eq. (10); Laguerre polynomials in this setting go back to Coffey (2005, 2007, 2010)
the arithmetic interpretation	Sekatskii , arXiv:1305.1421
asymptotics of the generalized sums as $n \rightarrow \infty$	Sekatskii , arXiv:1403.4484; and Voros , <i>Math. Phys. Anal. Geom.</i> 9 (2006) 53–63
the spurious exponential term $(1 + 1/b)^n$ and its cancellation	Sekatskii , arXiv:1404.7276, eq. (7) and §4
Li coefficients beyond ζ	Lagarias (automorphic L), Smajlović (Selberg class)
the <i>other</i> one-parameter family, $\rho \mapsto \rho/(\rho - \tau)$	Freitas (2006); Droll (2012) — the τ -Li coefficients
explicit detection indices: which n certifies a zero outside a region	Brown (2005); Bucur , Ernvall-Hytönen , Odžak & Smajlović (2016); Palojärvi (2019), whose Theorems 3.1 and 3.3 give explicit intervals $[n_0, n_1]$ with tables
numerical behaviour of τ -Li coefficients for functions that violate RH	Bucur , Ernvall-Hytönen , Odžak & Smajlović (2016)
the ongoing programme	Ernvall-Hytönen , Odžak , Smajlović & Zubača (2024), <i>Variants of the Li-Type Criteria</i> ; Voros (2020), discretized Keiper/Li
the shifted family carried to the Selberg class, with arithmetic and asymptotic representations	Mazhouda & Sodaïgui (2022), <i>The Li-Sekatskii coefficients for the Selberg class</i> — the name this paper adopts
the shifted criterion restated and extended by its author	Sekatskii (2021), in <i>Schrödinger Operators, Spectral Analysis and Number Theory</i> , Springer Proc. Math. Stat. 348

Nothing in §2 is new. It is written out because §3 needs it in one notation.

1.1 Two different one-parameter families, and which question each answers

There are two parameterised families of Li coefficients in circulation and they are **not** the same object.

	τ -Li: $\rho \mapsto \rho/(\rho - \tau)$	shifted Li: $\rho \mapsto (\rho - a)/(\rho - (1 - a))$
foci of the map	0 and τ	a and $1 - a$
the line it tests	$\text{Res} = \tau/2 - \mathbf{moves\ with}$ τ	$\text{Res} = 1/2 - \mathbf{fixed}$
what the parameter does	selects which half-plane is being certified	changes how the same line is viewed
what it is used for	zero-free regions, explicit $[n_0, n_1]$ certificates	—
source	Freitas 2006; Droll 2012	Sekatskii 2013

This paper is about the second family only. The question here — which shift best reveals a zero at a given height, with the tested line held at $1/2$ — is a detector question, not a zero-free-region question, and it is not the question the τ -literature answers.

[C] The genre, however, is established, and the paper should not be read as opening it. Explicit detection indices for Li-type coefficients are a developed subject: Brown (2005) related finitely many non-negative coefficients to zero-free regions; Palojärvi (2019) proved two-sided theorems giving explicit intervals $[n_0, n_1]$ such that a negative coefficient in that range certifies a zero outside a region, with numerical tables; and Bucur, Ernvall-Hytönen, Odžak and Smajlović (2016) studied the numerical behaviour of τ -Li coefficients for functions that violate RH — which is, numerically, the same kind of experiment as §3.3 here. Those results are rigorous with explicit constants; what follows is a closed-form optimisation and a measurement, and is much smaller in scope.

In particular, the heuristic used in §3.3 — that the crossing index scales inversely with the exponential rate — is the informal version of what Palojärvi’s Theorem 3.3 makes precise, her n_0 and n_1 depending on $\log R$. The estimate here is not a substitute for that and does not carry its constants.

[C] How far the prior-art search went. Four passes: the classical sources; the τ -Li and zero-free-region literature of 2005–2026; Sekatskii’s own continuation, which reached 2021 and gave the family its name; and a targeted sweep for parameter optimisation in any Li-type family, which returned nothing. Not consulted: MathSciNet or zbMATH, Droll’s thesis, and the full texts of Mazhouda-Sodaïgui (2022) and the Ernvall-Hytönen-Odžak-Smajlović-Zubača chapter (2024), all three of which are the most likely places for a statement of the optimum to be sitting. A reader who finds it there is asked to say so.

[O] That also names the one edit that would make this paper more than a measurement. Palojärvi’s argument produces explicit intervals $[n_0, n_1]$ for the τ family from the zero-counting bounds and a bound on the far-zero contribution. The same argument structure should carry to the Li-Sekatskii family, since the two differ only in where the foci sit; the far-zero estimate would be taken at q_a rather than $\rho/(\rho - \tau)$, and the rate $\text{artanh}(\beta/R)$ of Proposition 2 would enter where $\log R$ enters hers. Carrying that through would turn “the matched lens reaches the evidence at an index smaller by T ” from a scaling statement into a certified interval. It is not attempted here.

What is offered as new

One statement and one measurement, both in §3:

1. **[P]** within Sekatskii's shifted family, the closed form for the detection rate $\log |q_a(\rho)|$ of an off-line zero, its exact maximum at $d = |\rho - 1/2|$ with value $\operatorname{artanh}(\beta/|\rho - 1/2|)$, and the resulting factor- T gain over the classical lens. The optimisation of the shift against a target height does not appear in the sources of §1, but the surrounding genre does — see §1.1;
 2. **[M]** the numerical confirmation of that law, including the observation that a previously measured empirical improvement of $30.9\times$ for a planted zero at height 30 is the law's $T = 30$;
 3. **[P]** the two ingredients a Palojärvi-type detection interval needs, worked out for this family (§3.4): the on-line terms are $1 - \cos(n\theta)$ with $\theta = 2 \arctan(\gamma/d) - \pi$, and the far zeros contribute a positive $n^2 d^2$ term. The uniform estimates that would make it a certified interval are not supplied.
-

2. The construction [K]

2.1 The map

For real $a \neq 1/2$ the Möbius map $q_a(\rho) = (\rho - a)/(\rho - (1 - a))$ satisfies

$$|q_a(\rho)| = 1 \iff \operatorname{Re} \rho = 1/2,$$

because $|\rho - a| = |\rho - (1 - a)|$ says ρ is equidistant from the real points a and $1 - a$, whose perpendicular bisector is the line $\operatorname{Re} s = 1/2$. The functional equation gives $q_a(1 - \rho) = 1/q_a(\rho)$, so the image multiset is closed under $z \mapsto 1/z$ and **the zero sum must be read with symmetric summation**, never term by term.

2.2 The two objects

$$S_n(a) = \sum_{\rho} [1 - q_a(\rho)^n]$$

$$L_n(a) = 1/(n-1)! \cdot d^n/ds^n [(s-a)^{(n-1)} \log \xi(s)] \big|_{s=1-a}$$

At $a = 0$ these are Li's λ_n .

Proposition 1 [K]. $S_n(a) = (1 - 2a)L_n(a)$.

Verified. At $n = 1$ this is the Hadamard identity: $1 - q_a(\rho) = (2a - 1)/(\rho - 1 + a)$, so $S_1(a) = (2a-1) \sum_{\rho} 1/(\rho-1+a) = (1-2a) \cdot \xi'/\xi(1-a) = (1-2a)L_1(a)$. Numerically at $n = 1, 2$ and $a = 0, -0.5, -1.5$, the zero sum converges monotonically to the derivative side as the zero count grows through 100, 200, 400; at $a = 0, n = 1$ the target is 0.02309569, against the classical $\lambda_1 = 1 + \gamma/2 - \frac{1}{2} \log(4\pi) = 0.0230957089661$.

[C] A numerical trap in evaluating L_n . $\log \xi$ is regular at $s = 1$, but its natural expression is not: $\log(s-1)$ and $\log \zeta(s)$ are each singular there and cancel. Computing $(s-1)\zeta(s)$ as one quantity fixes the value but not the derivative, because an automatic differentiator's default step lands at $s - 1 \sim 10^{-40}$, where $\zeta \sim 10^{40}$ and the product

loses every digit. With *mpmath.diff* at default settings this returns -0.554 for λ_1 . An explicit step of 10^{-2} recovers 0.0230957089658, ten digits.

2.3 The prime side

For $\text{Res} > 1$ the derivative object is a generalized Laguerre transform of the von Mangoldt function,

$$S_{n^{\prime}}(a) = -(1 - 2a) \sum_{m \geq 2} \Lambda(m) m^{(a-1)} L_{n-1}^{(1-2a)}(\log m),$$

which is Sekatskii's Lemma 1.

[M] Checked here rather than quoted. Comparing the ζ -part of $L_n(a)$ against the sum directly, for $a < 0$ so that $1 - a > 1$ and the Dirichlet series converges, the two sides agree to the sign shown, and the residue is truncation rather than error:

relative residue $\ LHS +$ $RHS(M)\ /\ LHS\ $	$M = 500$	2000	8000	32000
$a = -0.5,$ $n = 1$	5.99×10^{-2}	3.04×10^{-2}	1.56×10^{-2}	8.14×10^{-3}
$a = -1.5,$ $n = 1$	3.32×10^{-4}	1.54×10^{-4}	1.32×10^{-4}	1.29×10^{-4}

At $a = -0.5$ the residue falls steadily with the prime cutoff. At $a = -1.5$ the sum converges fast and the residue settles on 1.3×10^{-4} , which is the finite-difference floor of the derivative side, not a disagreement.

The discriminating test is the behaviour in M , not the size at one M . A variant carrying an extra $1/\log m$ in the coefficient — an easy misreading of the source — gives residues that *stall* instead of converging:

variant residue	$M = 500$	2000	8000
$a = -0.5, n = 1$	3.700×10^{-1}	3.657×10^{-1}	3.639×10^{-1}
$a = -1.5, n = 1$	1.729×10^{-2}	1.732×10^{-2}	1.732×10^{-2}

At $a = -1.5$ the wrong coefficient is only 1.7% off and would pass a single-cutoff check; it is the flat column, against a correct residue falling to 1.3×10^{-4} , that excludes it.

The tail criterion $\limsup_n |S_n(a)| (1/n) \leq 1$ excludes off-line zeros, and obtaining it from this side requires cancellation inside the Laguerre sum of essentially RH strength.

3. The lens law

3.1 The rate

Proposition 2 [P]. Let $\rho = 1/2 + \beta + iT$ with $\beta > 0$, let $d = 1/2 - a$ be the lens width, and put $R = |\rho - 1/2| = \sqrt{(\beta^2 + T^2)}$. Then

$$\log |q_a(\rho)| = \frac{1}{2} \cdot \log[((\beta+d)^2 + T^2) / ((\beta-d)^2 + T^2)],$$

this has a unique stationary point in $d > 0$, at

$$d = R,$$

and it is a maximum, with value

$$\max_d \log |q_a(\rho)| = \frac{1}{2} \cdot \log[(R+\beta)/(R-\beta)] = \operatorname{artanh}(\beta/R).$$

Proof. Substituting $a = 1/2 - d$ gives $\rho - a = (\beta + d) + iT$ and $\rho - (1 - a) = (\beta - d) + iT$, which is the closed form. Write $N = (\beta + d)^2 + T^2$ and $D = (\beta - d)^2 + T^2$. Then $(N/D)' = 2[(\beta + d)D + (\beta - d)N]/D^2$, and with $u = \beta + d$, $v = \beta - d$ the bracket is $u(v^2 + T^2) + v(u^2 + T^2) = (u + v)(uv + T^2) = 2\beta(\beta^2 - d^2 + T^2)$. Since $\beta > 0$ this vanishes exactly at $d^2 = \beta^2 + T^2$, changing sign from $+$ to $-$. At $d = R$ one has $N = 2R(R + \beta)$ and $D = 2R(R - \beta)$, giving the stated value. ■

[M] Every line of that proof is machine-checked. `verify_li_lens.py` check 3a verifies the substitution, the form of $|q_a|^2$, the derivative's numerator, the factorisation through $u = \beta + d$, $v = \beta - d$, its collapse to $2\beta(\beta^2 - d^2 + T^2)$, the values of N and D at $d = R$, the uniqueness of the root and the sign change — and, since it is displayed below, every line of Sekatskii's route as well: fifteen symbolic identities, each with residual exactly zero, plus the sign change numerically. The final artanh step is checked as three separate algebraic facts rather than one transcendental identity, because $\operatorname{artanh}(z) = \frac{1}{2} \log((1+z)/(1-z))$ holds only for $|z| < 1$: the script verifies that N/D at $d = R$ is $(R + \beta)/(R - \beta)$, that $(x - 1)/(x + 1) = \beta/R$ for that x , and that $R^2 - \beta^2 = T^2 > 0$ so the hypothesis $\beta/R < 1$ is met. The domain condition was missing from the first draft of the proof and the check is what surfaced it.

[P] The same optimum in two variables — a derivation of S. K. Sekatskii. In correspondence (August 2026) Sekatskii gave a shorter route to the same result, which he offered as an exercise and which is reproduced here with his permission. It stays in σ and a where the argument above moves through five variables. Writing $\rho = \sigma + iT$,

$$|q_a(\rho)|^2 = \frac{(\sigma - a)^2 + T^2}{(\sigma + a - 1)^2 + T^2} = 1 + \frac{(2\sigma - 1)(1 - 2a)}{D}, \quad D = (\sigma + a - 1)^2 + T^2,$$

and setting the derivative in a to zero gives

$$a = \frac{1}{2} \pm \sqrt{\varepsilon^2 + T^2}, \quad \varepsilon = \sigma - \frac{1}{2},$$

so $d = \frac{1}{2} - a$ has $|d| = \sqrt{\varepsilon^2 + T^2} = R$: the same optimum, with his ε the β used here. The displayed second line uses $(\sigma + a - 1)^2 - (\sigma - a)^2 = (2\sigma - 1)(2a - 1)$;

Sekatskii notes that he did not track the overall sign, which does not affect where the extremum sits. The two derivations were checked against each other symbolically and agree exactly.

[C] And his own reading of what it is worth. Sekatskii holds that the maximum of the modulus, and hence the choice of a , **might** be — rather than is — important for numerical work. That qualification is adopted here. Nothing below claims the optimum matters in practice; §5 says the opposite, and the measurements of §3.3 are of a planted configuration in a finite truncation.

The geometric reading. The map q_a has its two foci at a and $1 - a$, placed symmetrically about $1/2$ at distance d . The proposition says the detector is sharpest when **the foci stand at the same distance from $1/2$ as the zero does.**

Corollary 3 [P]. The gain over the classical lens is exactly

$$G(\beta, T) = \operatorname{artanh}(\beta/R) / \frac{1}{2} \log[((\beta+\frac{1}{2})^2 + T^2) / ((\beta-\frac{1}{2})^2 + T^2)], \quad R = \sqrt{(\beta^2 + T^2)},$$

and, as $\beta \rightarrow 0$ with T fixed, the numerator is $\beta/T + O(\beta^3/T^3)$ and the denominator is $\beta/(\frac{1}{4} + T^2) + O(\beta^3)$, so

$$G \rightarrow \frac{\frac{1}{4} + T^2}{T} = T + \frac{1}{4T}.$$

The excess $1/(4T)$ is not rounding. It is the $\frac{1}{4}$ carried by the classical lens $d = 1/2$, and it is exactly what the gain column above measures: 10.025, 30.008, 100.0025, 500.0005 against the printed 10.026, 30.009, 100.003, 500.001, the remaining difference being the $O(\beta^2)$ term at $\beta = 0.1$. Dropping it and writing $G \rightarrow T$ would be a scaling statement, not the limit.

Since the index at which an off-line zero first drives the sum negative scales inversely with the exponential rate, **the detection threshold improves by a factor of T .**

3.2 Measured

[M] The maximiser, located by direct search over $d \in [0.05, 5T]$ at 4×10^5 points:

T, β	10, 0.1	30, 0.1	100, 0.1	30, 0.01	500, 0.01
argmax_d	10.00049	30.00011	100.00013	29.99993	500.00000
$\sqrt{(\beta^2 + T^2)}$	10.00050	30.00017	100.00005	30.00000	500.00000
max rate	9.999833e-3	3.333327e-3	9.999998e-4	3.333333e-4	2.000000e-5
$\operatorname{artanh}(\beta/R)$	9.999833e-3	3.333327e-3	9.999998e-4	3.333333e-4	2.000000e-5

[M] The gain, as the ratio of rates:

T	10	30	100	500
rate at $d = T$	1.000×10^{-2}	3.333×10^{-3}	1.000×10^{-3}	2.000×10^{-4}
rate at $d = 1/2$	9.974×10^{-4}	1.111×10^{-4}	1.000×10^{-5}	4.000×10^{-7}

T	10	30	100	500
gain	10.026	30.009	100.003	500.001

with $\beta = 0.1$; the $\beta = 0.01$ row gives the same gains to three decimals, as the $O(\beta^3)$ term requires.

3.3 A previously measured improvement, explained

[M] An earlier experiment planted a symmetric quartet at $\rho = 1/2 \pm 0.1 \pm 30i$ among verified critical-line zeros and located the first negative index: $n \approx 41646$ at $a = 0$, and $n \approx 1348$ with a lens of width $d \approx 30$, an improvement of $30.89\times$. Proposition 2 predicts the ratio of rates,

$$(3.333327 \times 10^{-3}) / (1.110790 \times 10^{-4}) = 30.01,$$

against the measured 30.89.

[M] The 3% residue was afterwards measured and is scatter, not a systematic effect. The detection index was swept over 24 runs — six heights $T = 10, 20, 30, 50, 80, 120$, two displacements $\beta = 0.1, 0.05$ and two ordinate counts 2000, 8000 — and the measured gain divided by T has mean 1.0034 with standard deviation 0.0163 and slope -0.0052 against $\log T$. The departures **shrink** with T rather than growing, and move when only the zero count changes. An earlier draft of this note guessed that the 3% was systematic and came from the background's own dependence on d ; that guess is withdrawn. The crossing index is still not exactly inversely proportional to the rate — the on-line zeros contribute a background growing like $(n/2) \log n$ (Sekatskii, arXiv:1404.7276, Theorem 6) rather than a constant — but at these heights that correction is smaller than the run-to-run scatter. Regenerated by code/detection_index_sweep.py.

The empirical 30.9 was the height of the planted zero.

3.4 Why the threshold scales like $1/\log R$, and what is still missing

§3.3 used the rule that the crossing index scales inversely with the rate. That rule is not assumed here; the two ingredients Palojärvi's argument needs are available for this family too.

Proposition 4 [P]. For a zero **on** the critical line, $\rho = 1/2 + i\gamma$, and lens width d ,

$q_a(\rho) = e^{i\theta}$, $\theta = 2 \arctan(\gamma/d) - \pi$, so $\operatorname{Re}(1 - q_a(\rho)^n) = 1 - \cos(n\theta) \geq 0$, and $\theta = -2 \arctan(d/\gamma) = -2d/\gamma + O(d^3/\gamma^3)$. Hence for $\gamma \gg nd$ the term is $2n^2d^2/\gamma^2 + O(\cdot)$.

Checked: $|q_a| - 1 = 1.1 \times 10^{-16}$ on the line, and $|\operatorname{Re}(1 - q_a^n) - (1 - \cos n\theta)| \leq 1.7 \times 10^{-14}$ over $d = 0.5, 30$, $\gamma = 50, 500, 5000$, $n = 3, 100$.

[M] The far zeros therefore contribute a positive term growing like n^2d^2 . Summing $2n^2d^2/\gamma^2$ against the Riemann-von Mangoldt density gives

$$\Sigma_{\{|\gamma| > T\}} \operatorname{Re}(1 - q_a^n) \approx (2n^2 d^2 / \pi) \cdot (\log(T/2\pi) + 1) / T,$$

measured against the exact sum over 2×10^6 ordinates:

d, n	$T = 5 \times 10^3$	2×10^4	10^5
0.5, 20	0.993	0.974	0.892
0.5, 50	0.993	0.974	0.892
30, 50	—	—	0.892

as ratios of exact to model.

Two features of that table are not typographical. The first two rows are identical because the factor $n^2 d^2$ appears in the exact sum and in the model alike and cancels from the ratio, so the ratio does **not** depend on d or n at all — which is itself a check on the model, and the reason the third row is given at one T only. The dashes are the condition $\gamma \gg nd$: at $d = 30, n = 50$ the scale $nd = 1500$ is not small against a cut at $T = 5 \times 10^3$, so those two cells would be testing the expansion rather than the density.

The shortfall at $T = 10^5$ is the finite zero list — the model integrates to infinity while the sum stops. The shipped script keeps the cut below $\gamma_{\max}/10$ so the table adapts to whatever list is supplied, **which means the numbers a reader sees depend on the list they have.** With a 100,000-ordinate list reaching $\gamma = 74921$ the cut falls to $T = 5 \times 10^3$ and the ratio is 0.910 rather than 0.993; the shape is the same and the larger shortfall is the shorter list. The script prints the list's reach next to the table so the two are never confused.

The structure of the argument then closes. The far term is the analogue of Palojärvi's $K_{F,3}(T, \tau)n(n-1)$; for the near zeros, an off-line zero with $|q_a(\rho)| \geq R$ gives, by the same lemma of Montgomery she uses,

$$\Sigma_{\{|\gamma| \leq T\}} \operatorname{Re}(1 - q_a^n) < N(T) - (1/20) R^n$$

for some n in an explicit range. Combining, $S_n(a)$ is driven negative once

$$R^n \gtrsim 20 [N(T) + (2n^2 d^2 / \pi) (\log(T/2\pi) + 1) / T], \quad \text{i.e. } n \gtrsim \log(\text{background}) / \log R.$$

So the detection index is inversely proportional to $\log R$ up to a logarithmic correction, and Proposition 2 — which maximises $\log R$ over the lens — minimises it. That is the derivation §3.3 was missing.

[C] What this is not. It is the argument's *structure*, not a theorem. Palojärvi's intervals come with explicit constants obtained from uniform estimates on every term, and none of that uniformity is established here: the $O(d^3/\gamma^3)$ remainder is not bounded uniformly in n , the range of n for which Montgomery's lemma applies is not made explicit, and the background's own dependence on d — which enters through $n^2 d^2$ — is not carried through. A draft of this note guessed that the last omission was where the 3% residue of §3.3 lives; the 24-run sweep reported there withdraws that guess. Supplying the uniform estimates would turn the factor- T gain from a scaling law into a certified interval, and is the concrete next step named in §1.1.

4. The truncation artifact [K]

[C] Truncating the prime-Laguerre sum at $m \leq M$ produces apparent exponential growth in n even when no off-line zero is present. Measured n -th root growth: 3 at $a = -0.5$, $5/3$ at $a = -1.5$.

These are not empirical constants. With $d = 1/2 - a$,

$$(d + 1/2)/(d - 1/2) = (a-1)/a = (b+1)/b, \quad b = -a,$$

giving 3, $5/3$, $9/7$ at $a = -0.5, -1.5, -3.5$ — exactly the term $(1 + 1/b)^n$ in Sekatskii's eq. (7), which he traces to the formal zero of ξ at $s = 1$ and shows is cancelled by the pole of ζ once the prime sum is treated to first order.

So the artifact has a known analytic source and a known cancellation. Raising the cutoff postpones it and does not remove it; only doing the regrouping before truncating does. Any numerical search for off-line zeros through this criterion must subtract this term first, or it will find one.

5. What the lens does not buy

The lens law is a statement about **detector sensitivity**, not about provability. Three things follow, and none of them is a route to a proof.

1. **The criterion is unchanged.** For every admissible a , non-negativity of $S_n(a)$ for all n is equivalent to RH — that is Sekatskii's theorem. Choosing a well changes which n first reveals a violation; it does not change what has to be proved.
2. **The hard direction is untouched.** Establishing $\limsup |S_n(a)|^{1/n} \leq 1$ from the prime side requires cancellation in the Laguerre transform of von Mangoldt of essentially RH strength, at every a . Moving the lens changes the arithmetic weights m^{a-1} and the polynomial argument $(1 - 2a) \log m$; it does not make the cancellation easier to exhibit.
3. **The gain is bounded by what one is looking for.** A factor of T helps only if one already knows roughly where to look. Scanning T costs back what matching to T gains.

Where it is genuinely useful: as a targeted test. Given a specific height at which one suspects a violation, the matched lens reaches the same evidence at an index smaller by T , which at $T \sim 10^3$ is the difference between a feasible computation and an infeasible one. That is a numerical instrument, and it is worth having as one.

A more sensitive detector is not a shorter proof.

6. The same question for the Weil form, and why the answer differs

The shifted Li criterion is not the only finite window on the explicit formula. The truncated Weil quadratic form — Connes-van Suijlekom, and the Galerkin truncations studied numerically by Groskin — is another, and Sekatskii notes the connection outright: his arithmetic interpretation is obtained from Weil's explicit formula, and he

cites Bombieri's *Remarks on Weil's quadratic functional*. So the two constructions are two finite windows on one identity, and it is fair to ask the same question of both.

In the Weil setting the free parameter is the support length L . A test function supported in $[-L/2, L/2]$ has a transform that extends to an entire function, and an off-line zero at $\rho = 1/2 + \beta + iT$ is probed at the complex point $T - i\beta$.

[K] By Paley-Wiener, for any f supported in $[-L/2, L/2]$,

$$|\hat{f}(T - i\beta)| \leq e^{(\beta L/2)} \cdot \|f\|_1,$$

and the factor is sharp. So the greatest amplification the support length permits is $\beta L/2$ in the log.

[P] That ceiling has no interior maximum. $\beta L/2$ is strictly increasing in L . There is no optimal support length; longer is always better.

[M] The Dirichlet sine modes sit comfortably inside the ceiling and do not saturate it: at $T = 30$, the ratio $|\hat{\varphi}_j(T - i\beta)|/(\|\varphi_j\|_1 e^{\beta L/2})$ for the mode nearest resonance is

L	4.5	9	18	36
$\beta = 0.1$	0.632	0.517	0.363	0.210
$\beta = 0.5$	0.312	0.173	0.087	0.044

against ceilings $e^{\beta L/2}$ of 1.25, 1.57, 2.46, 6.05 and 3.08, 9.49, 90.0, 8103.

The contrast

	shifted Li	truncated Weil
free parameter	lens width $d = 1/2 - a$	support length L
detector rate	$\text{artanh}(\beta/R)$, $R = \rho - 1/2 $	ceiling $\beta L/2$
behaviour in the parameter	interior maximum at $d = R$	monotone, no maximum
cost of moving it	none — a is free	on the CvS path $L = \log c$, so a longer support forces a larger prime sum

[C] The two rows are not the same kind of object and should not be read as one. The Li entry is the *actual* rate for a given zero, determined by the parameter alone. The Weil entry is a *ceiling* over all test functions of that support, not the rate of any particular one. The comparison is between what each parameter can buy, not between two measured rates.

The structural reading. The Li parameter is a *position*: q_a has foci at a and $1 - a$, and there is a matching condition because a position can be right or wrong relative to the target. The Weil parameter is a *bandwidth*: more is monotonically better, and it is bounded not by an optimum but by a price. On the coupled path that price is the prime

sum, which is exactly the coupling $L = \log c$ that Groskin identifies as structural for the reference construction.

[O] Whether the Weil form has an optimum in a parameter that is a position — the prime cutoff held independent of the support, say — is not settled here. Off the saturated diagonal that family is strongly indefinite, so “detection” would first have to be given a meaning that does not reduce to prime deletion.

7. Reproducibility

```
python code/verify_li_lens.py           # Propositions 1-4, the prime formula, and the ta
python code/detection_index.py          # the detection index at a few heights
python code/detection_index_sweep.py    # the 24-run gain table of §3.3
```

All three read ZETA_ZEROS — a file of critical-line ordinates, one per line, increasing. `verify_li_lens.py` uses it for check 3b alone and skips that one table without it, printing the cut and the list’s reach beside every number it does print; `detection_index.py` and `detection_index_sweep.py` fall back to `mpmath.zetazero`, which is far slower, and say so. A suitable list can be taken from Odlyzko’s tables (www.dtc.umn.edu/~odlyzko/zeta_tables) or the LMFDB. `detection_index_sweep.py` needs 8000 ordinates and exits nonzero if the mean gain departs from T by more than 2% or if the slope against $\log T$ turns positive.

§2.2 needs an explicit differentiation step (see the caveat there); the script sets it and fails loudly if the recovered λ_1 does not match the closed form. Tested with Python 3.12, NumPy 2.4, SymPy 1.14, mpmath 1.3.

AI assistance

The verification script and much of the prose in this note were written with the assistance of Claude (Anthropic); the research direction, the decisions about what to publish, and responsibility for every claim are the author’s. See the repository README for a fuller statement.

Acknowledgements

I am grateful to **S. K. Sekatskii** for reading an early version of this note and replying in detail; for the derivation reproduced in §3.1, which he offered as an exercise and permitted me to include; and for the qualification recorded there on what the optimum is and is not worth in numerical practice. Any error that remains is mine.

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No claim is made about the Riemann Hypothesis. The numerical results are exploratory measurements of a detector, not evidence about the location of any zero.